HDOC Day-59

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Batch	15

Problem Statement

Write a C program that reads a string **s**. Pass the string to a function named **longest_unique_substring()** using its address. Use pointers to pass the string between the calling function and **longest_unique_substring()**.

The function must find the longest substring containing no repeated characters and display its length. Uppercase and lowercase letters are treated as different characters. If the input string is empty, display **0**.

C Program

```
#include <stdio.h>
#include <string.h>

void longest_unique_substring(char *s)
{
    int last[256];
    int i, start = 0, maxLength = 0;

    for (i = 0; i < 256; i++)
        last[i] = -1;

    for (i = 0; *(s + i) != '\0'; i++)
    {
        unsigned char ch = *(s + i);

        if (last[ch] >= start)
            start = last[ch] + 1;

        last[ch] = i;

        if (i - start + 1 > maxLength)
            maxLength = i - start + 1;
    }

    printf("Length of longest substring: %d\n", maxLength);
}

int main()
{
    char s[1000];

    printf("Enter string: ");
    fgets(s, sizeof(s), stdin);

    s[strcspn(s, "\n")] = '\0';

    longest_unique_substring(s);

    return 0;
}
```

Explanation

The string is passed to **longest_unique_substring()** through its address. The array **last[]** stores the most recent position of each character. The variable **start** marks the beginning of the current substring with unique characters. When a repeated character appears inside the current substring, **start** moves to the position after its previous occurrence. The maximum length is updated while scanning the string once. Since characters are tracked separately, uppercase and lowercase letters are treated as different

characters.

Sample Outputs

```
Example 1
String: abcabcbb
Length of longest substring: 3

Example 2
String: bbbbb
Length of longest substring: 1

Example 3
String: pwwkew
Length of longest substring: 3

Example 4
String: [empty]
Length of longest substring: 0
```

Result

Thus, the program successfully finds the length of the longest substring without repeating characters using pointers and correctly displays 0 for an empty string.